

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
8	(a)			One bacterium will give rise to one colony/ 22 colonies on agar so 22 bacteria in original sample (1)	1			1		1
	(b)	(i)		36.5- 37.5 °C (1) human pathogen grows better at body temperature (1)			2	2		
		(ii)		Any two (×1) from: • Sterilisation of {dishes/agar} / use of {autoclave/ pressure cooker} for agar (1) • Glass spreader sterilised/ flamed (1) • Petri dish lid not raised too high (1) • <u>Lids</u> secured with <u>tape</u> (1) • Flame mouth of test tube (1) • Working close to bunsen burner (1)	2			2		2
	(c)			$7.3 \times 10^4 = 3$ marks (use of standard form) If incorrect award 2 marks for Number of colonies/cm ³ = $22 \div 0.0003 = 73\,333.33^\circ$ $3 \times 10^{-4} \text{ cm}^3$ 73.3×10^3 7.3×10^x 7.6×10^4 (used 23 colonies instead of 22) If incorrect award 1 mark $0.3 \div 1\,000 = 0.0003 = 1$ mark Number of colonies/cm ³ = $23 \div 0.0003 = 76\,666.66^\circ$ (used 23 colonies instead of 22)		3		3	3	3
	(d)	(i)		Any value within range $>3\% \leq 4\%$ (1)			1	1		1
		(ii)		Use (a range of) concentrations between 3 % and 4 % (1)			1	1		1
	(e)			Test on cells/ tissues (1) Test on (named) animals (1) Test on <u>healthy</u> {volunteers/ people} (1)	3			3		
				Total for Question 8	6	3	4	13	3	8